*Homework 5-6, Probability*

1) Let X be uniformly distributed over $[0, 1]$.
a) What's the probability that $X \leq t$ for a given $t$? (Suppose $0 < t < 1$.)
b) What's the probability that $X \leq t^2$ for a given $t$?
c) What's the probability that $\sqrt(X) \leq t$ for a given t?
d) Based on the above, write down the (cumulative) distribution function of $\sqrt(X)$.
e) Find the density function of $Y = \sqrt(X)$ (by differentiating the above).

2) Let X be uniformly distributed over $[0, \frac{\pi}{2}]$.
Find the distribution function and the density function of $Y = sinX$.

3) Waiting time at a dentist (expressed in hours) can be regarded as an exponentially distributed random variable, with parameter 2.
a) What's the expected waiting time?
b) What's the probability that we have to wait less than half an hour?
c) What's the probability that we have to wait more than half an hour, but less than 1 hour?
(The exponential distribution function with parameter $\lambda$ is $F(x) = 1 - e^{-\lambda x}$ and the density function is $f(x) = \lambda e^{-\lambda x}$, its expectation is $\frac{1}{\lambda}$ and its variance is $\frac{1}{\lambda^2}$.)

4) Lifetime of a light bulb can be modelled with Exponential distribution.
a) What's the parameter of this distribution, if we know that the average lifetime is 2.5 years?
b) What's the probability that a light bulb (with life time having this distribution) serves longer than 10 years?

5) Compare the following probabilities and put them into order. (Hint: these probabilities only depend on how many standard deviations above the mean the given value is.)
a) $P(X \leq 9)$, if $X \sim N(5, 16)$ (the notation means: the distribution of X is Normal (Gaussian) with mean 5 and variance 16).
b) $P(Y \leq 9)$, if $Y \sim N(-1, 100)$.
c) $P(Z \leq 9)$, if $Z \sim N(3, 9)$.
d) $P(V \leq 3)$, if $V \sim N(0, 3)$.

6) a) If $X$ has distribution $N(0, 4)$ then what's the mean and variance of $X/2$?
b) If $Y$ has distribution $N(3, 25)$ then what's the mean and variance of $Y - 3$?
c) What transformation would change $Y - 3$ in such a way that the variance becomes 1?

7) The Normal distribution with mean 0 and variance 1 is called Standard Normal distribution. Its distribution function ($\Phi(x)$) can not be given in an explicit form, but it can be given in a table with a certain precision (see for example http://www.thestudentroom.co.uk/attachment.php?attachmentid=134337&d=1330530156).
Some important values from the table: $\Phi(1.96) = 0.975$, $\Phi(1) = 0.8413$ , $\Phi(0) = 0.5$, etc.
Convert the following probabilities to such a form that you can look them up in the Normal distribution table, and find them.
a) $X \sim N(5, 16)$, $P(X \leq 9) =?$
b) $Y \sim N(-1, 100)$, $P(Y \leq 9) =?$
c) $X \sim N(5, 16)$, $P(X \leq 1) =?$
d) $Z \sim N(11, 4)$, $P(Z \leq 14) =?$
e) $V \sim N(-2.5, 25)$, $P(V \leq 10) =?$

8) Let $X$ and $Y$ be independent, identically distributed random variables, with common distribution function $F$ and density function $f$. Show that $V = max\{X, Y\}$ has distribution function $P(V \leq t) = F(t)^2$, and density function $f_V(t) = 2f(t)F(t)$. Find the distribution function and the density function of $U = min\{X, Y\}$.

9) I am selling my house, and have decided to accept the first offer exceeding 0.5 million Pounds. Assuming that offers are independent random variables with common Exponential distribution, with parameter 2.5 (calculating everything in million pounds), find the expected number of offers received before I sell the house.

10) a) Randomly choosing 2 points ($X$ and $Y$) in the interval $[0, 1]$, these points divide the interval into three parts. What's the probability that a triangle can be constructed from these three sections? Suppose both $X$ and $Y$ are uniformly distributed over $[0, 1]$, independently of each other.

(Hint: the possible pairs $(X, Y)$ can be plotted on the plane, in a coordinate system, and the possible $(X, Y)$ pairs form a square. Those pairs where a triangle can be constructed from the sections, form a smaller shape, whose probability is proportional to its area. This is a common method - called geometric solution.)

If you find this question easy, you can solve one of the following alternatives instead (for extra points):

b) What's the probability that an acute angled triangle can be constructed? (a triangle with all angles $< 90$ degree)

c) Buffon's needle: A plane is ruled by the lines $y = n(n = 0, +1, -1, +2, -2, ...)$ and a needle of unit length is cast randomly on to the plane. What is the probability that it intersects some line? We suppose that the needle shows no preference for position or direction.

(Buffon designed the experiment in order to estimate the numerical value of $\pi$. Try it if you have time!)